CODEFORCES
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A. Count the Inversion

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Here is a Pseudocode of the Merge Sort Algorithm.

```
def merge(a, b):
    # write your code here
    # a and b are two sorted list
    # merge function will return a sorted list after merging a and b

def mergeSort(arr):
    if len(arr) <= 1:
        return arr
    else:
        mid = len(arr)//2
        a1 = mergeSort(............)  # write the parameter
        a2 = mergeSort(............)  # write the parameter
        return merge(a1, a2)          # complete the merge function above
```

Now, you are given an array ***A*** of size ***N*** of ***N*** distinct integers. It is guaranteed that the array A contains a permutation of integers from 1 to N (i.e., every integer from 1 to N appears exactly once).

1. Count the number of inversions in the given array.
2. Sort the array in non-decreasing order.

An inversion is a pair (i, j) where $i < j$ and $A[i] > A[j]$.

Input

The first line contains an integer ***N*** $(1 \leq N \leq 10^5)$ — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq N)$ separated by spaces.

Output

In the first line, print the total number of inversions in the given array. In the next line, print the array in non-decreasing order.

Examples

input	Copy
5 1 2 5 4 3	

output	Copy
3 1 2 3 4 5	

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input Copy

```
5
1 2 3 4 5
```

output Copy

```
0
1 2 3 4 5
```

input Copy

```
5
5 4 3 2 1
```

output Copy

```
10
1 2 3 4 5
```

input Copy

```
7
6 4 2 5 7 3 1
```

output Copy

```
14
1 2 3 4 5 6 7
```

Note

In the first example (1-based indexing), the inversions are the pairs of indices $(3,\ 4), (3,\ 5)$ and $(4,\ 5)$.

In the second example, there are no inversions.

In the third example, every pair of i, j where $i < j$, we have $A[i] > A[j]$. Hence, All 10 such pairs are inversions.

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